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Williams

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(54) **FOLDING FIREARM**

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(21) Appl. No.: **17/957,554**

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(22) Filed: **Sep. 30, 2022**

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Primary Examiner — Bret Hayes

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Langlotz Patent & Trademark Works, LLC

(60) Provisional application No. 62/960,230, filed on Jan. 13, 2020.

(57) **ABSTRACT**

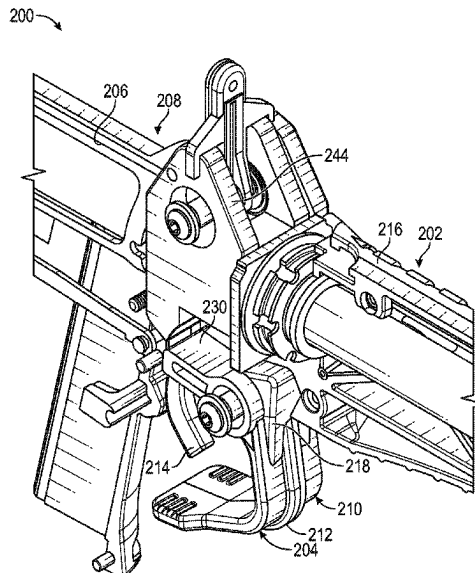
(51) **Int. Cl.**
F41A 11/04 (2006.01)

Folding firearms have a frame having a trigger assembly and a bolt passage, a barrel assembly pivotally connected to the frame for movement between an extended operable condition and a folded inoperable condition, the barrel assembly defining a barrel axis, a latch assembly operably connected to the frame and to the barrel assembly, the latch assembly including a latch operable between a first latched condition and a second unlatched condition, when the barrel assembly is in the operable condition and the latch is in the latched condition, the barrel assembly is secured in the operable condition, when the latch is in the unlatched condition, the barrel assembly is free to move away from the operable condition toward the folded condition, and the firearm being free of restraints on movement of the barrel assembly from the operable condition besides the latch, such that single-control release of the barrel assembly is provided.

(52) **U.S. Cl.**
CPC **F41A 11/04** (2013.01)

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CPC F41A 11/04; F41A 21/30; F41A 21/48;
F41G 1/02; F41G 1/06; F41G 1/16; F41G
1/17
USPC 42/71.02
See application file for complete search history.

38 Claims, 19 Drawing Sheets



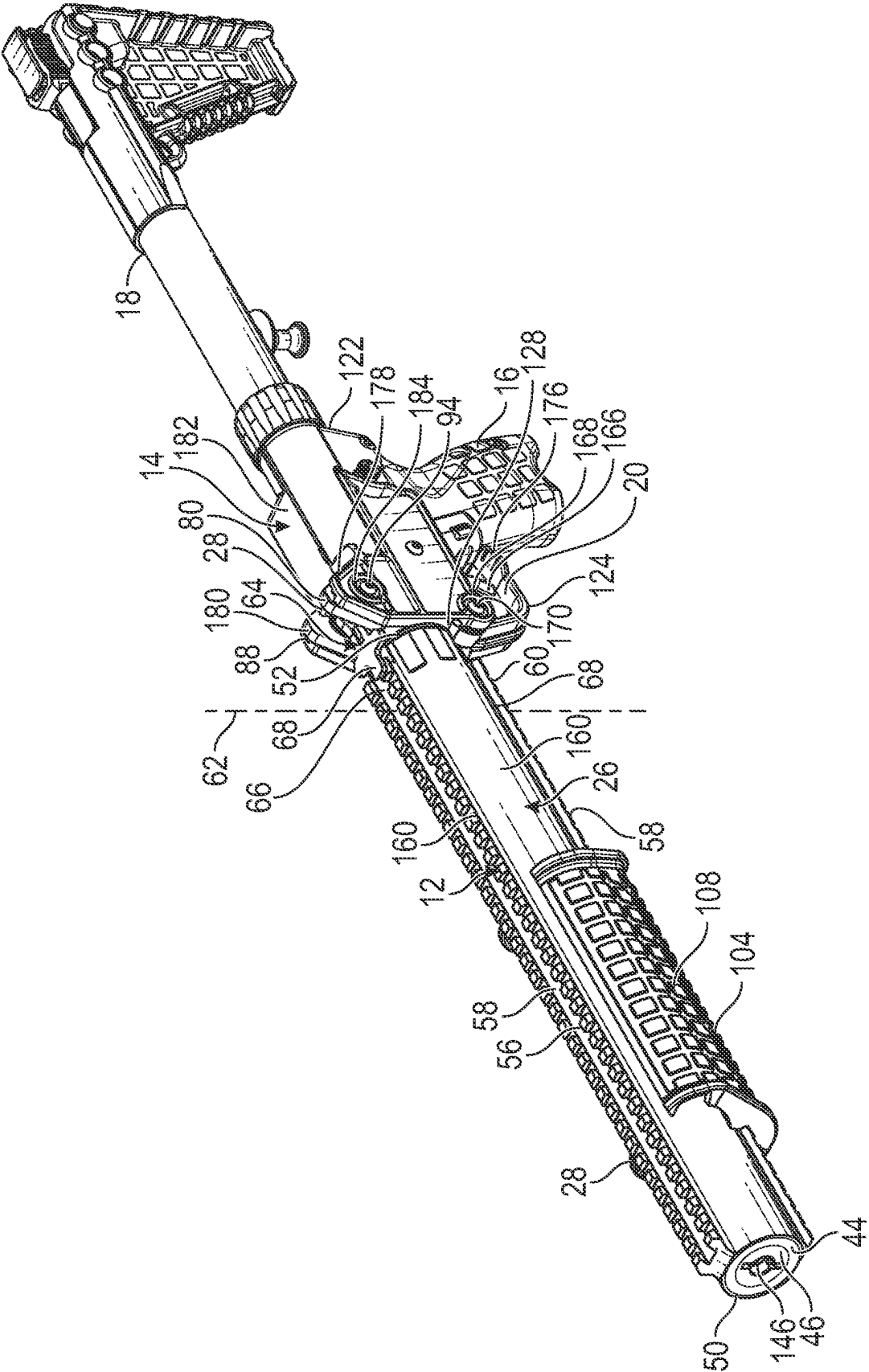


FIG. 1

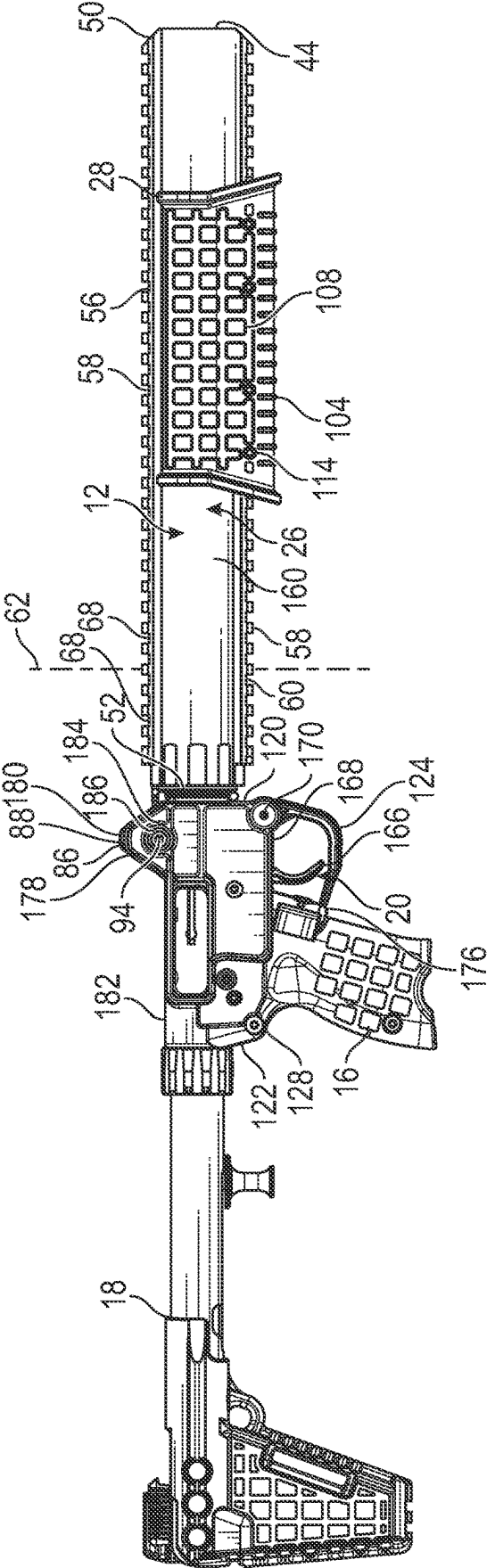
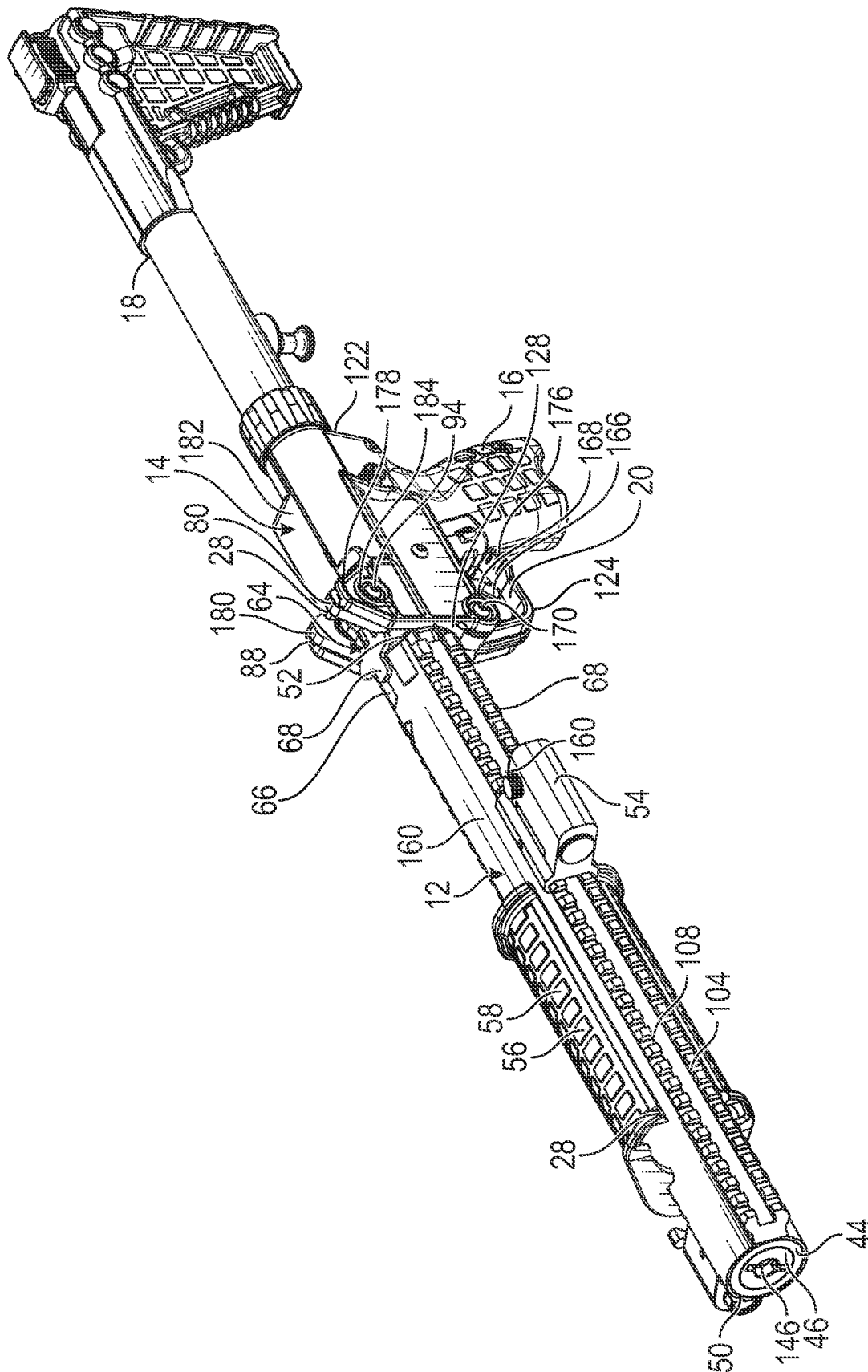


FIG. 2



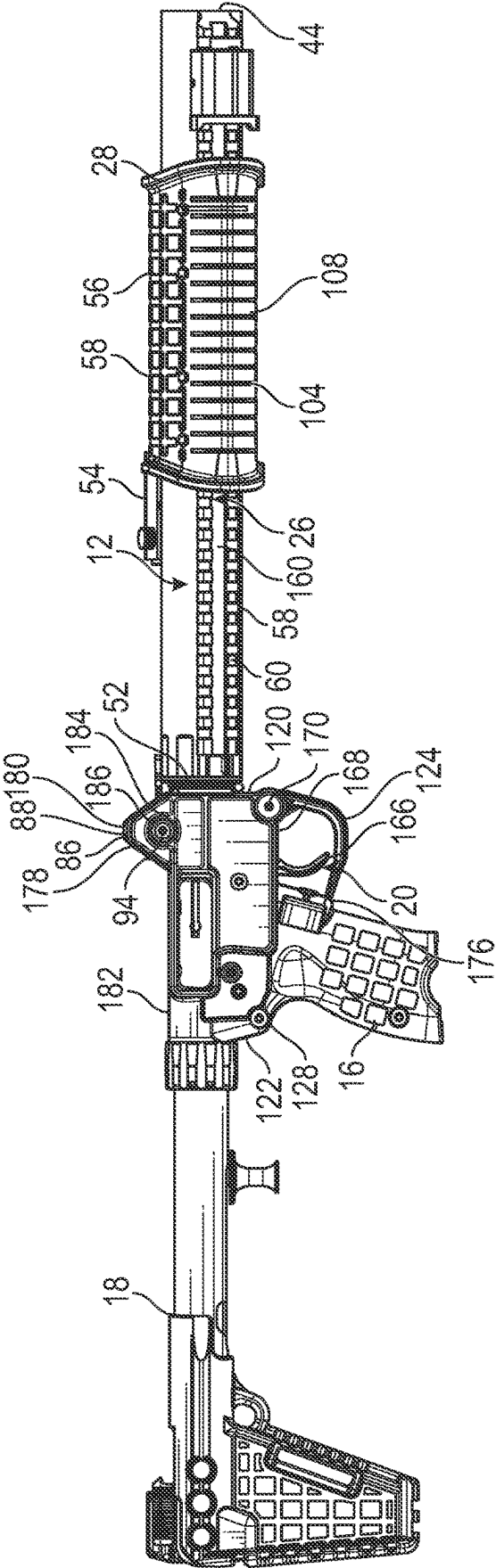


FIG. 4

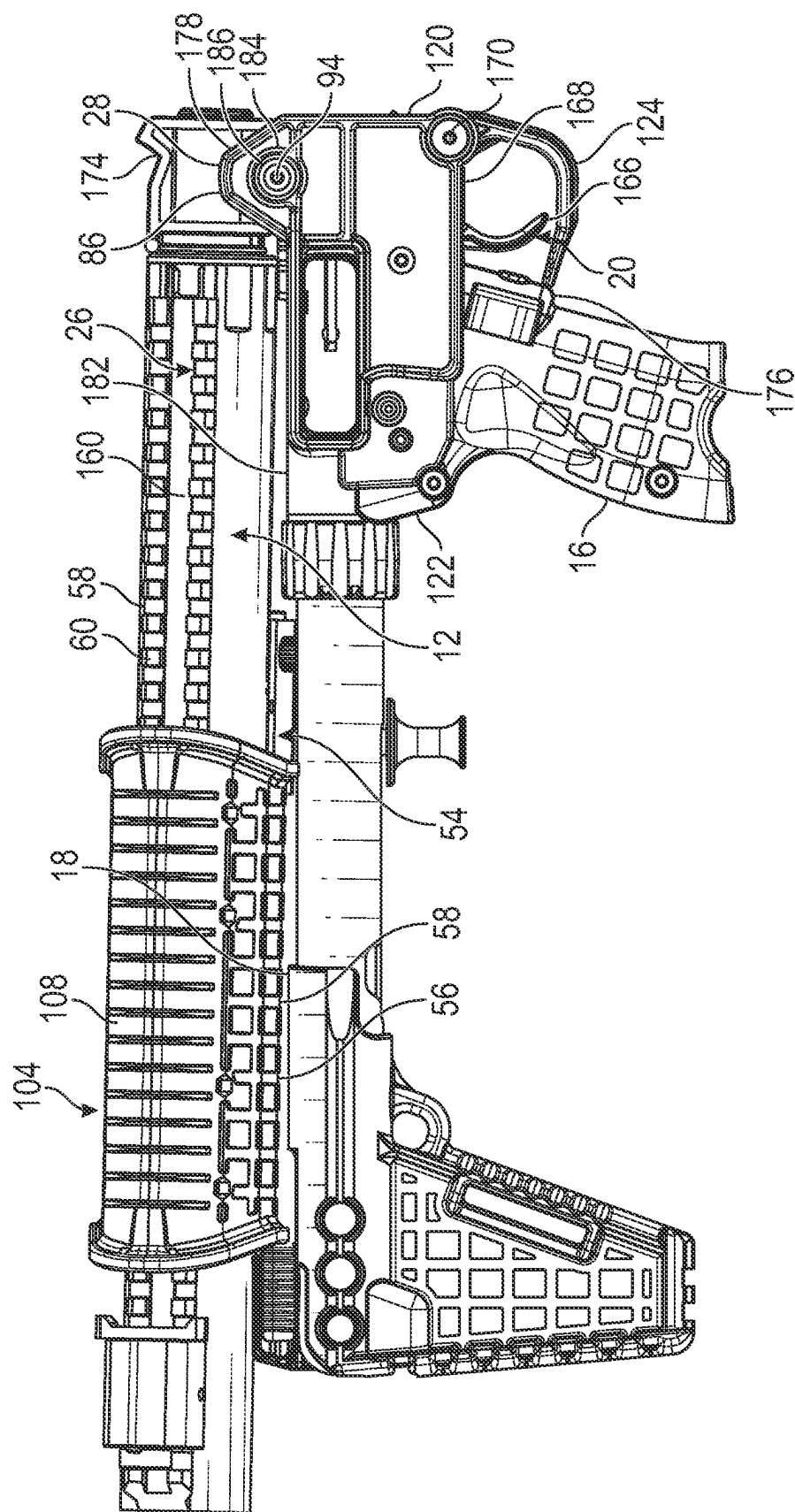


FIG. 5

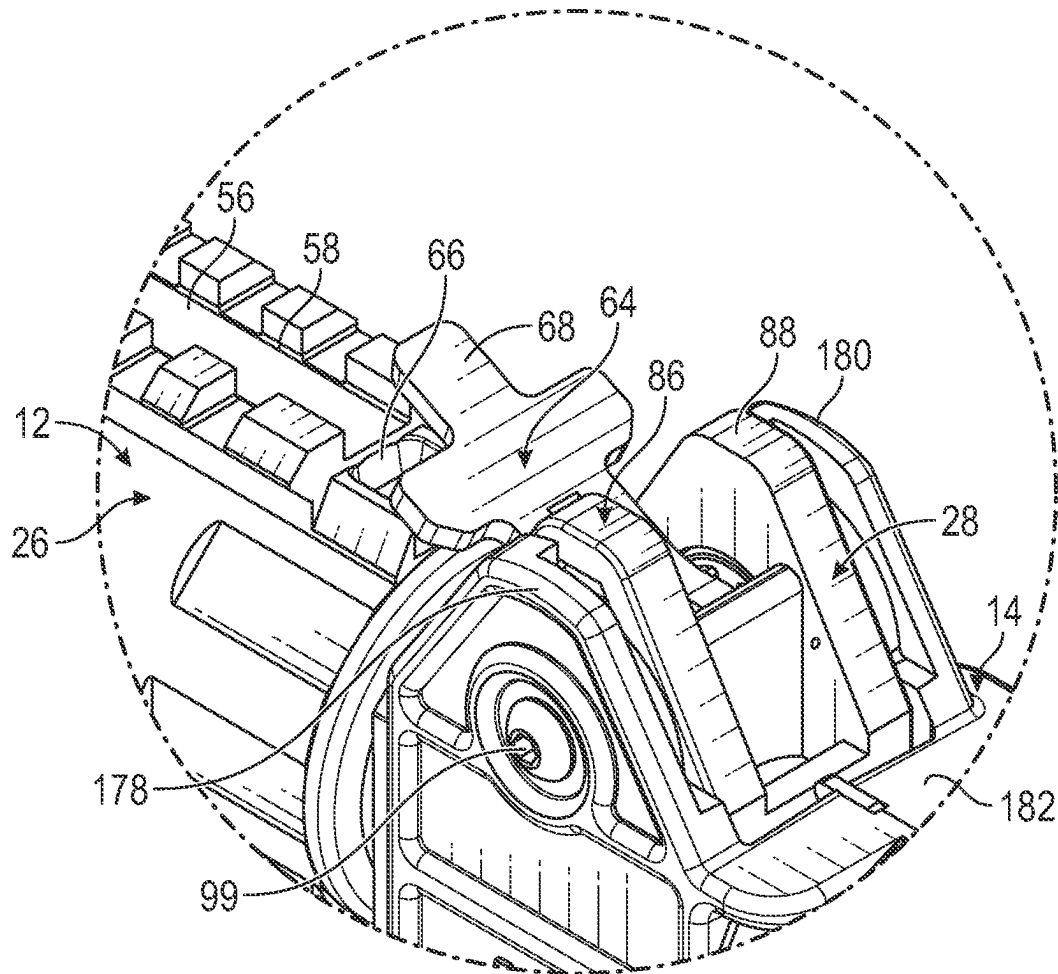


FIG. 6

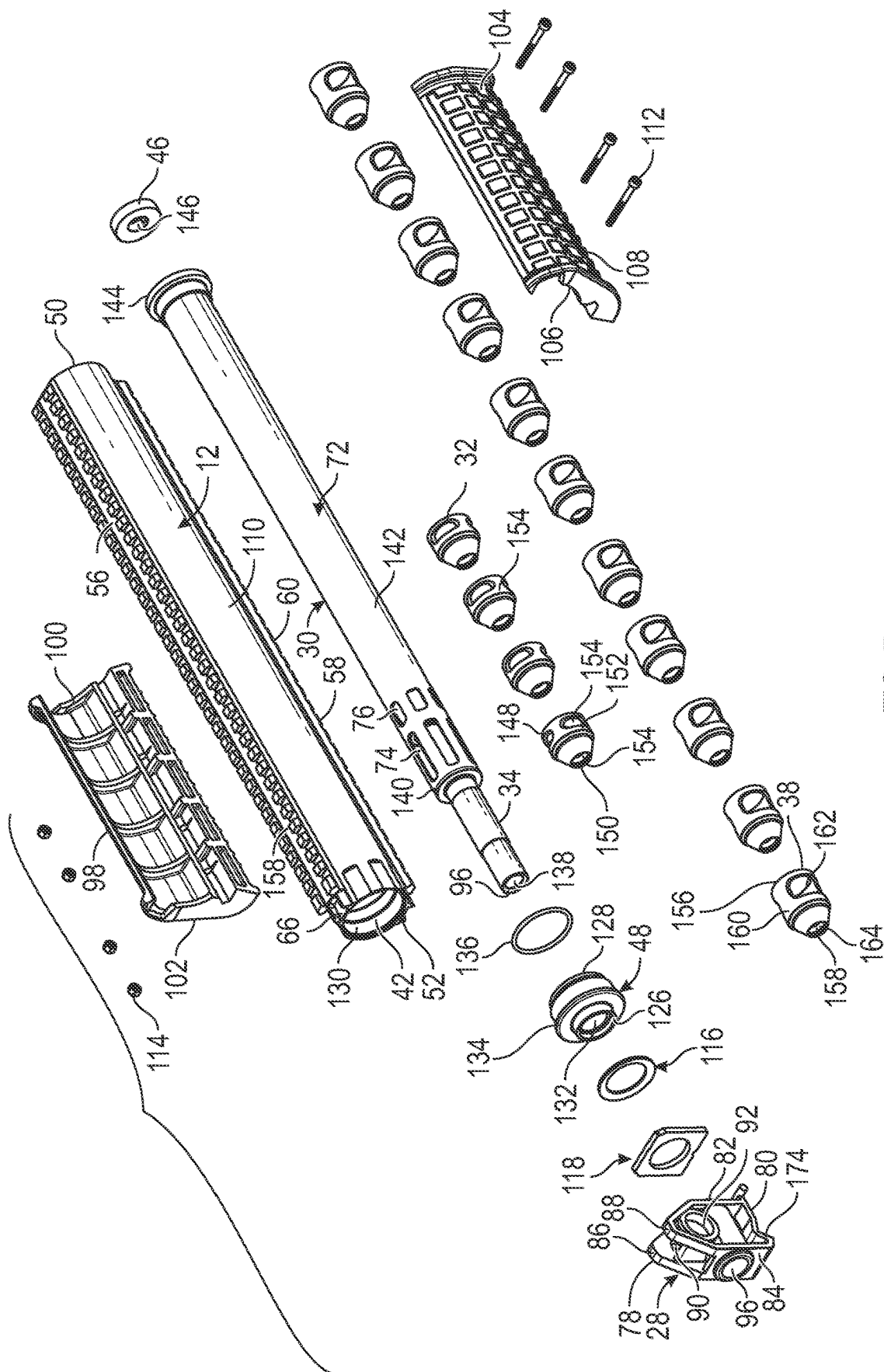
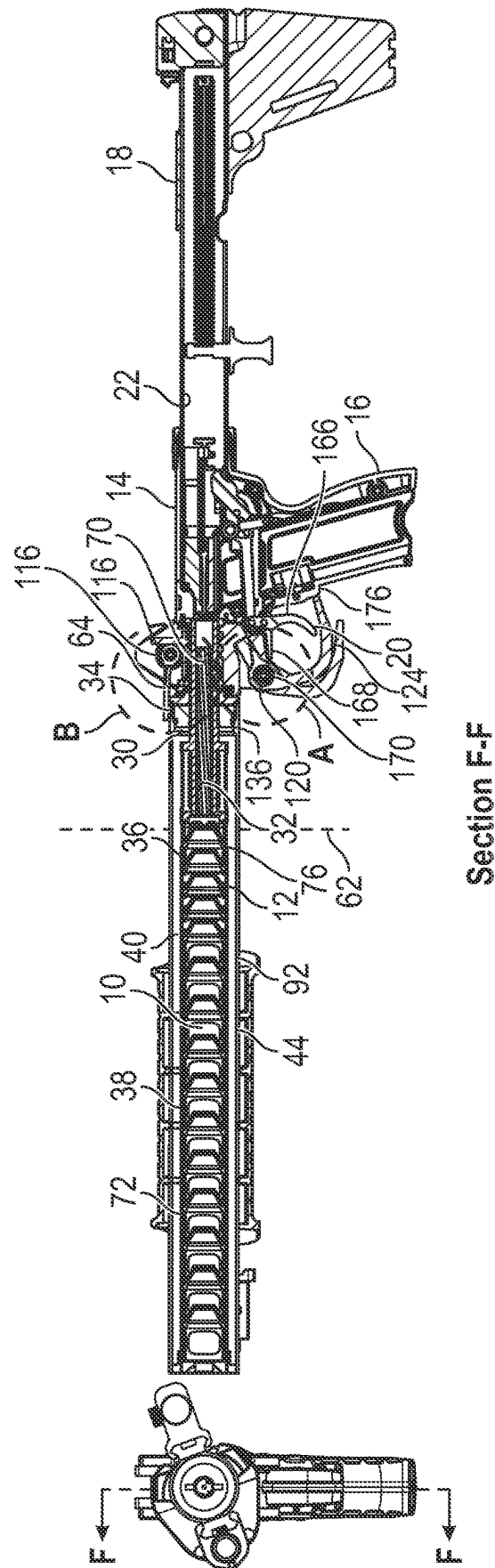
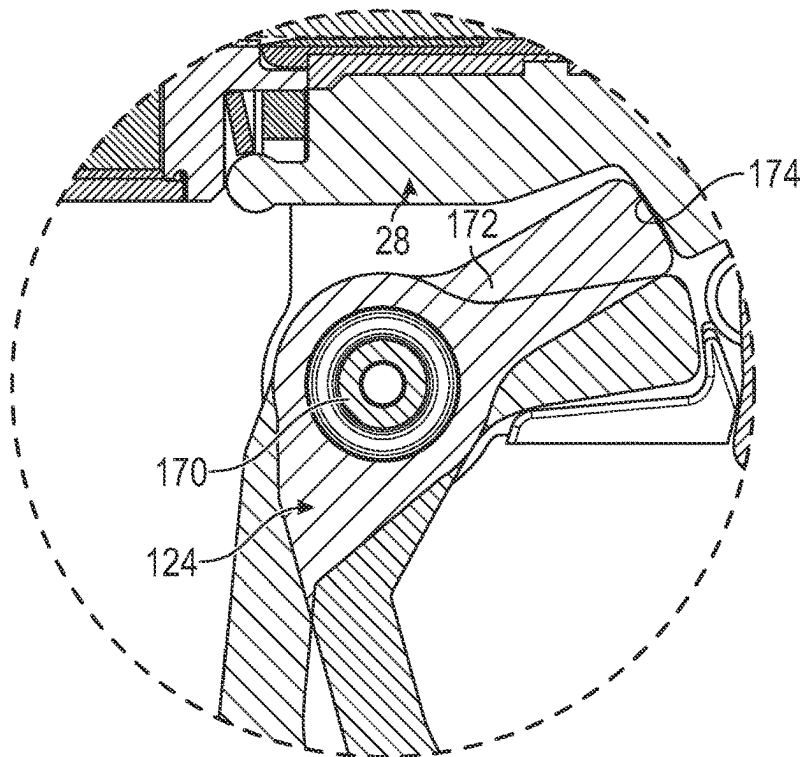


FIG. 7

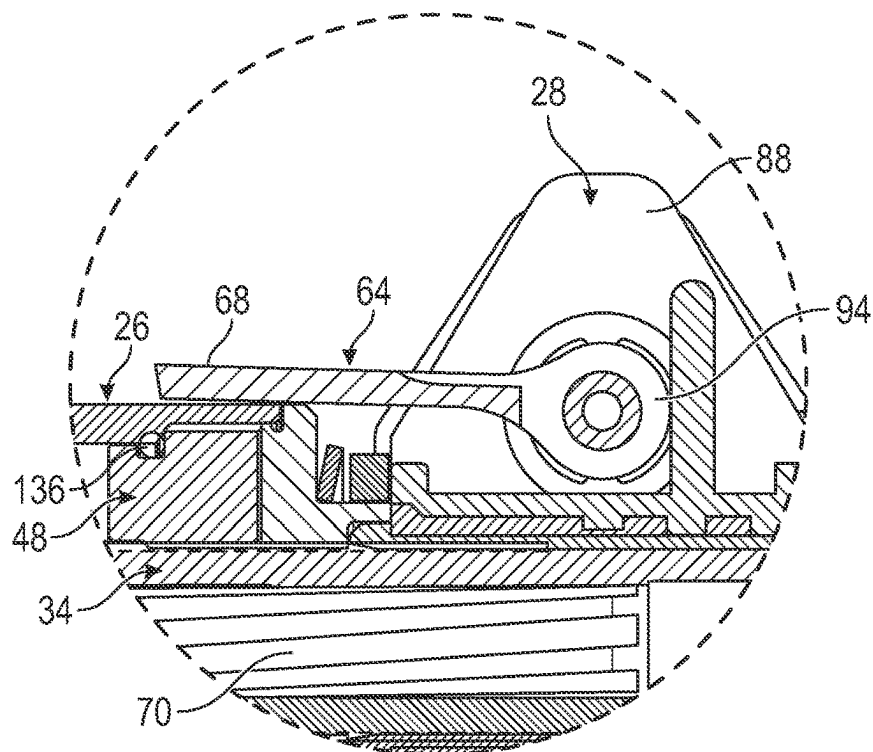


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Detail A

FIG. 9



Detail B

FIG. 10

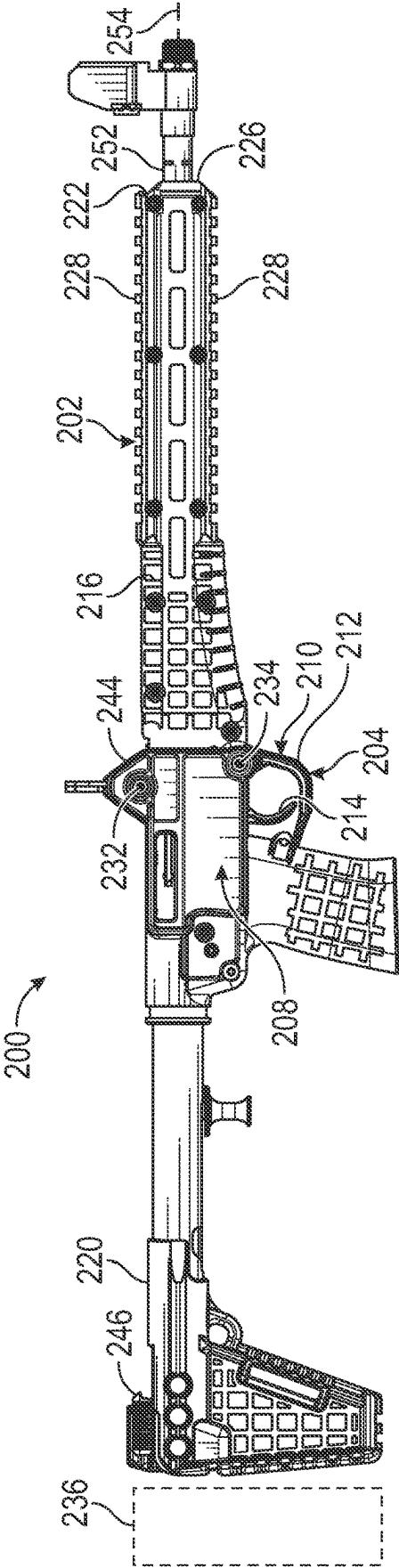


FIG. 11

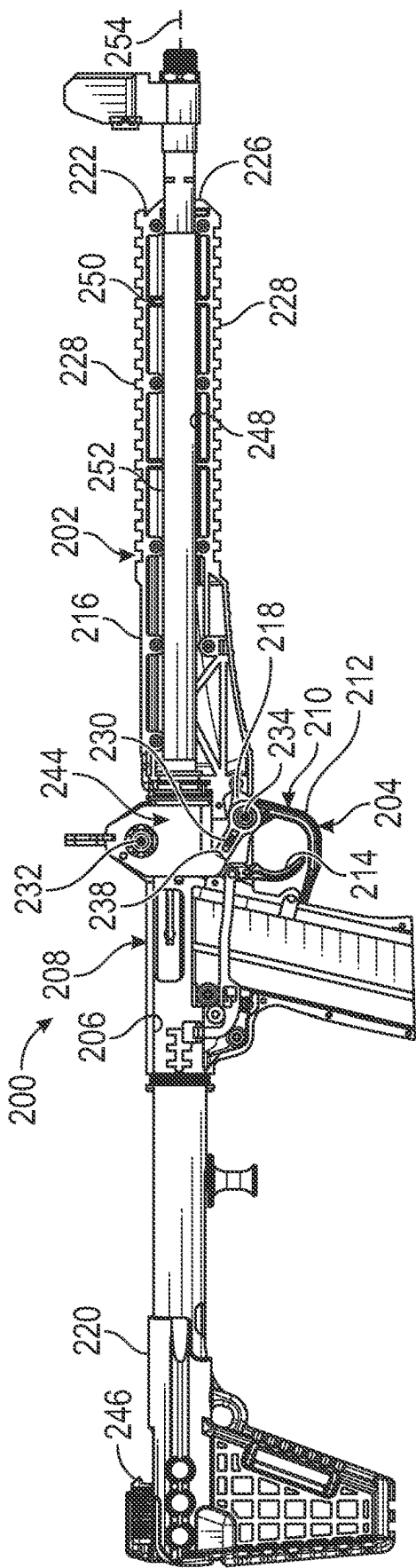


FIG. 12A

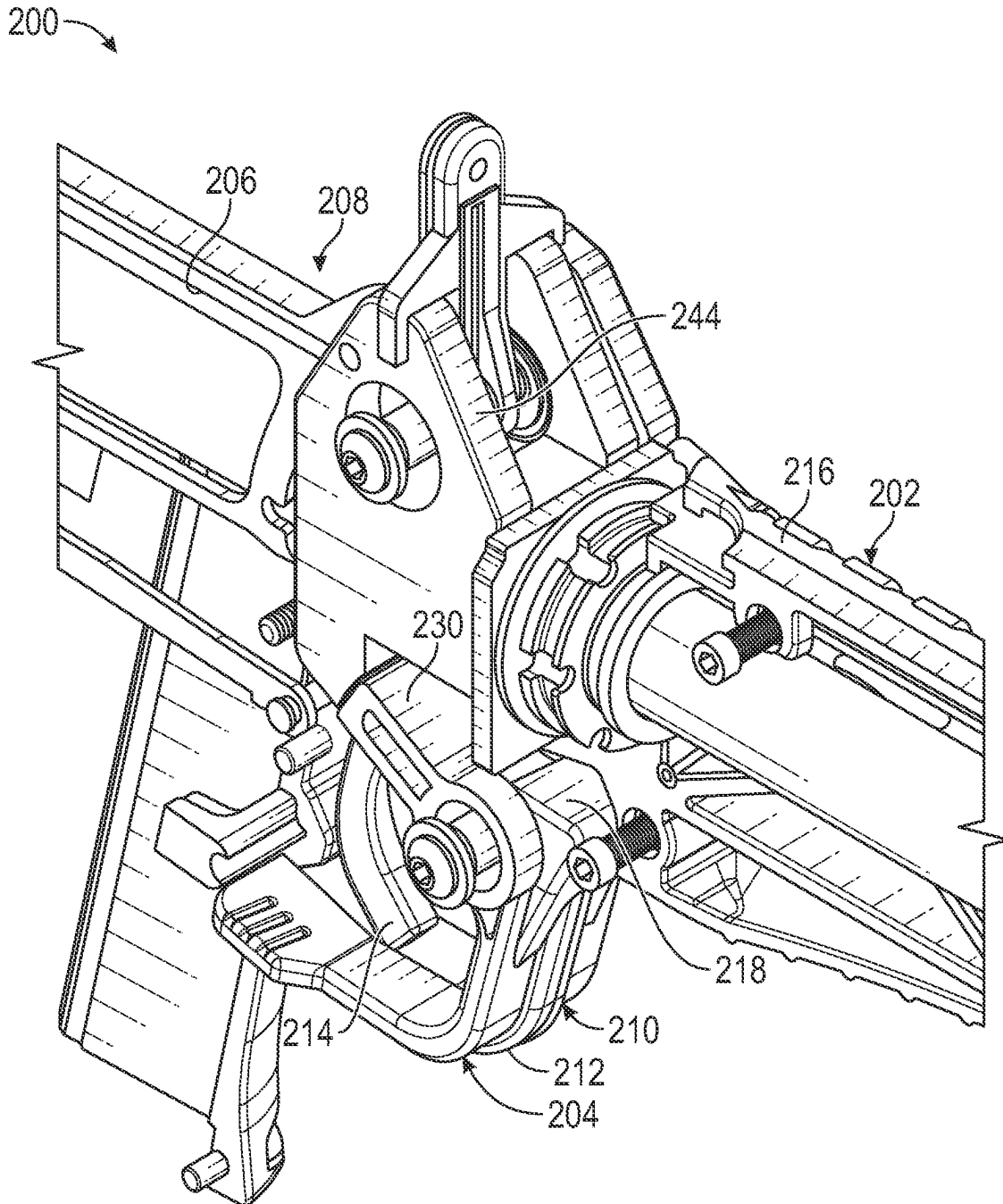
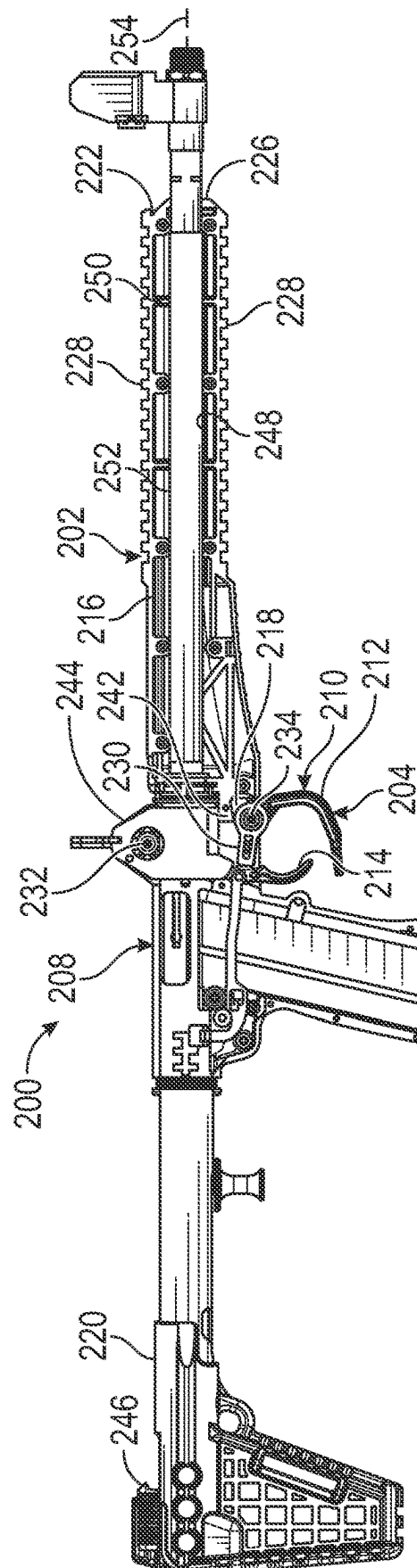


FIG. 12B



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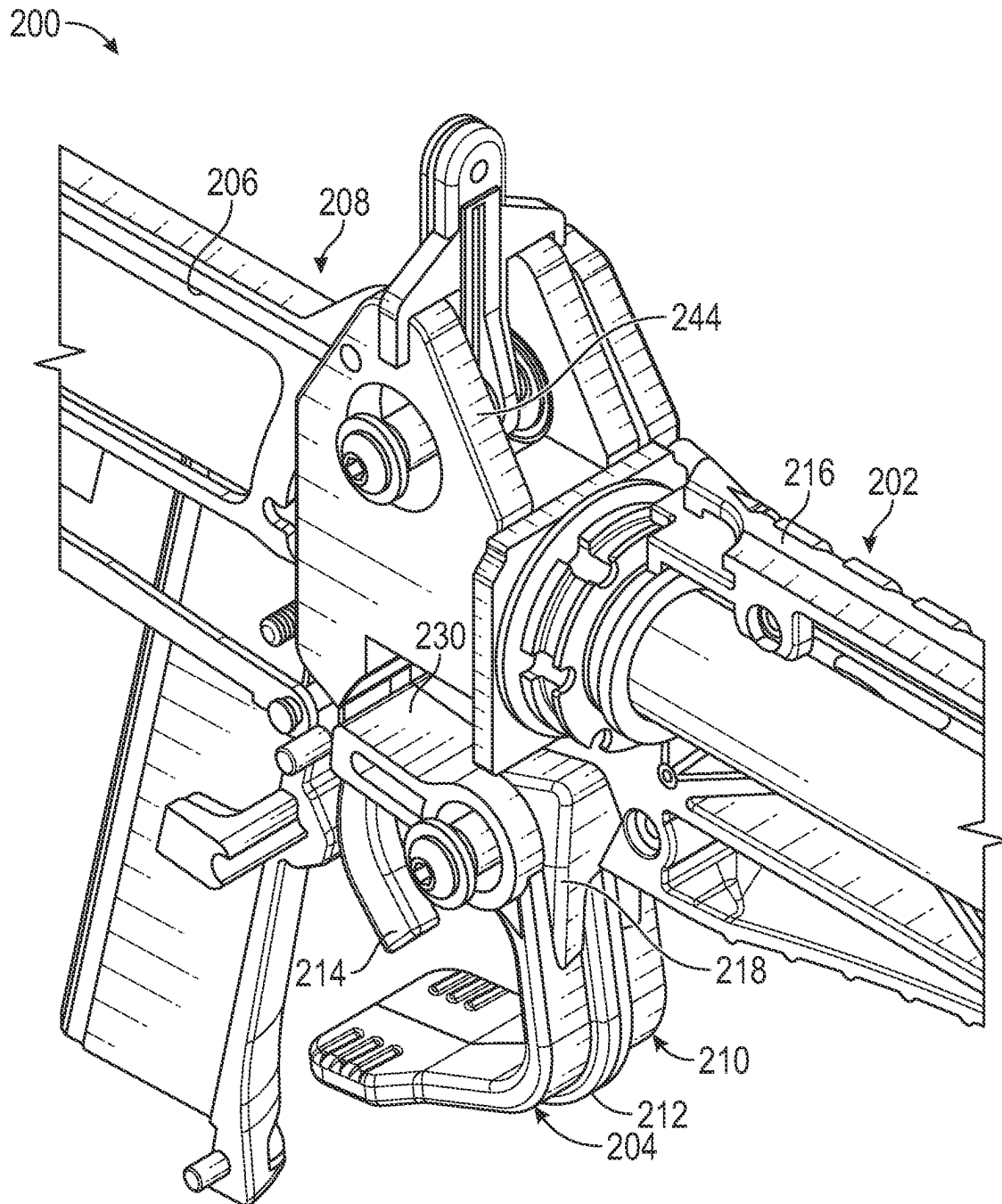


FIG. 13B

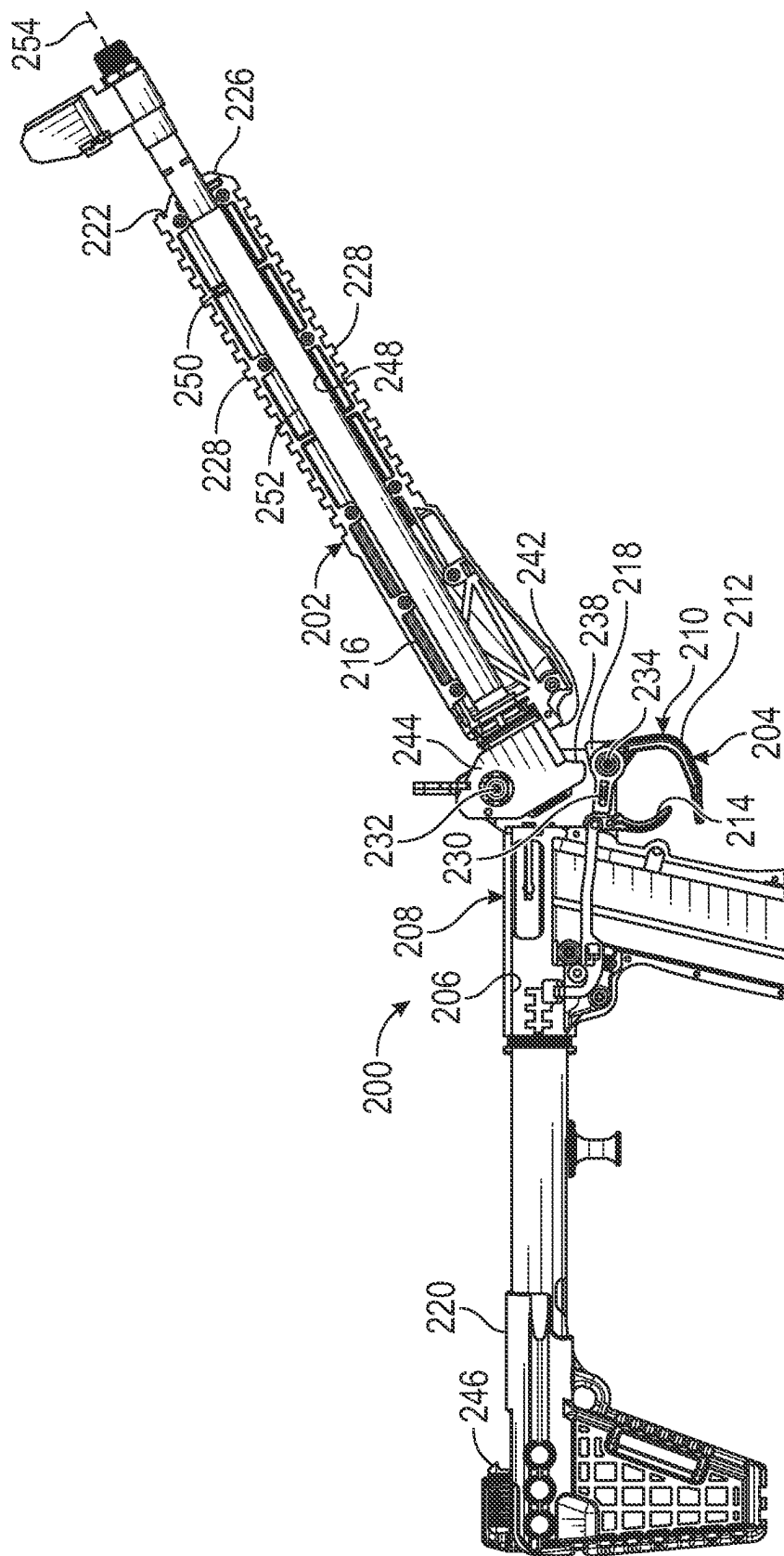


FIG. 14A

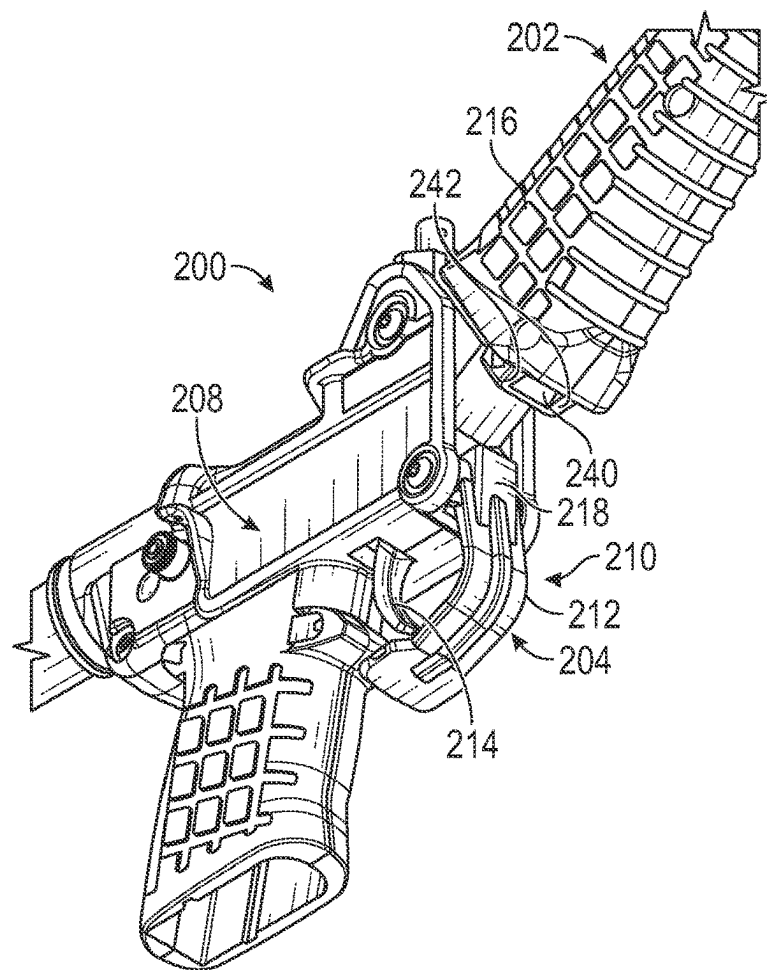


FIG. 14B

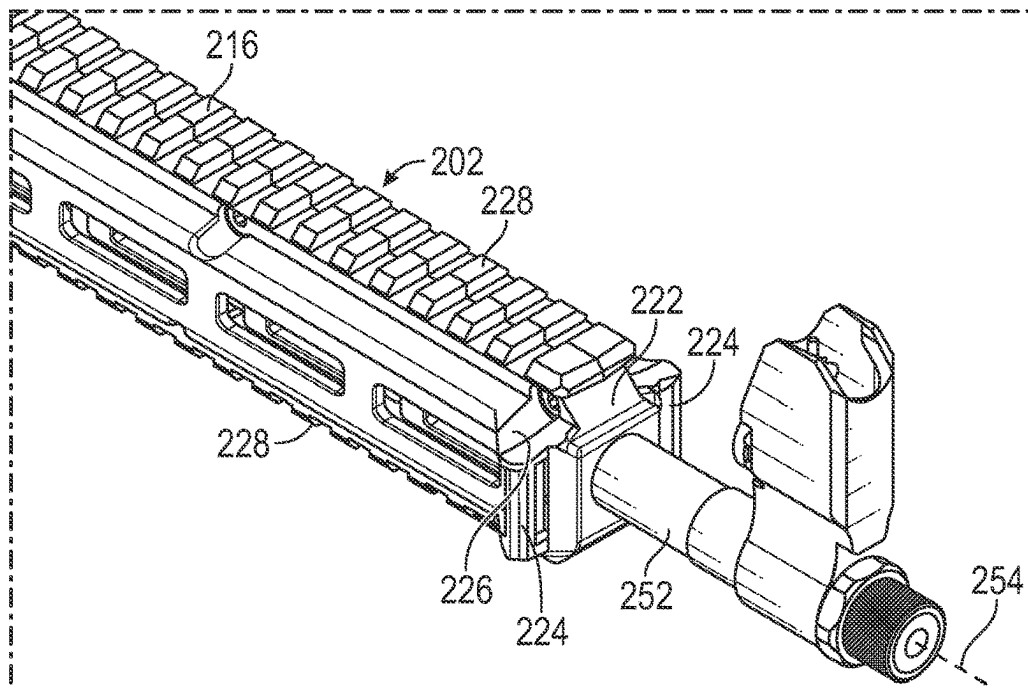


FIG. 15

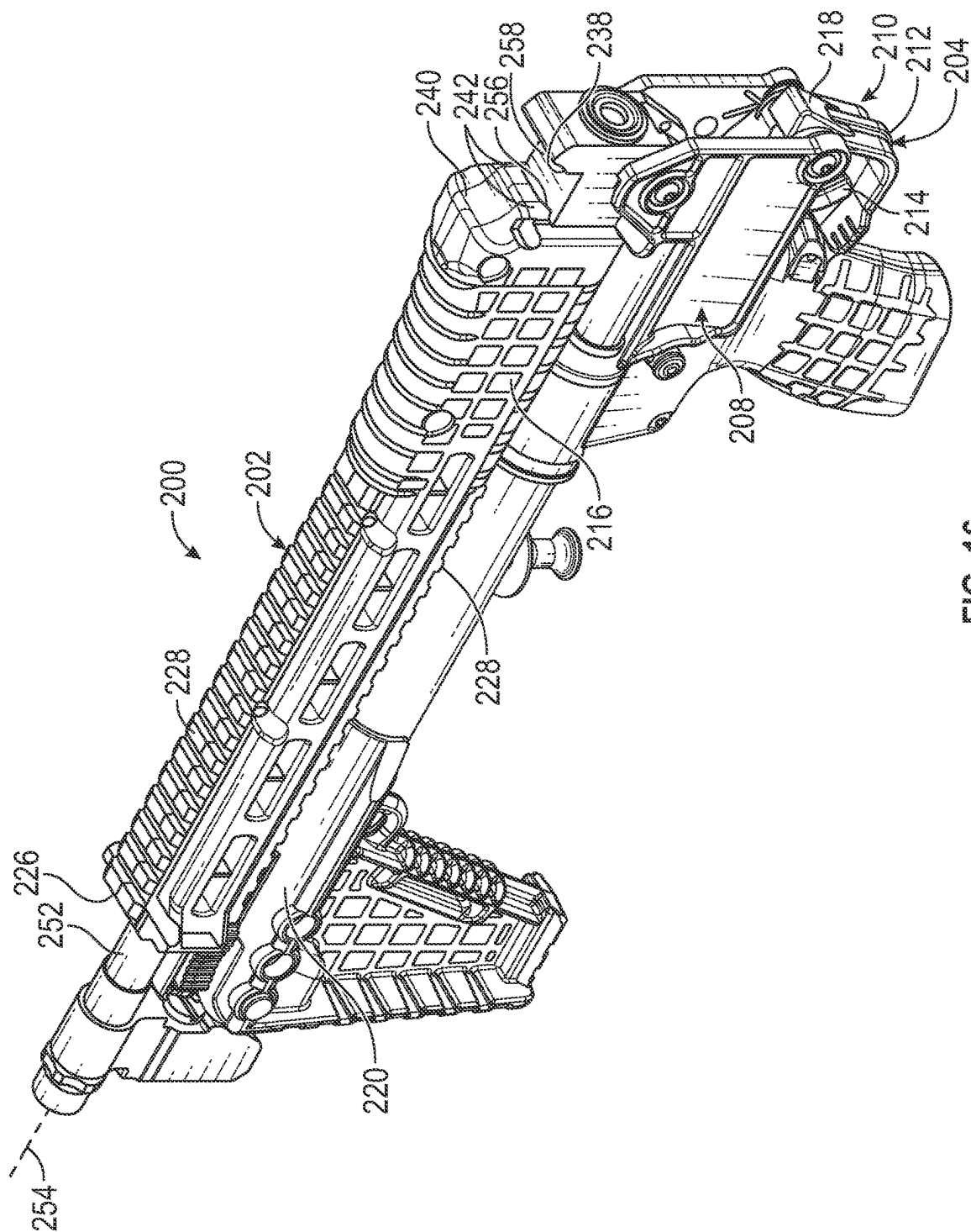


FIG. 16

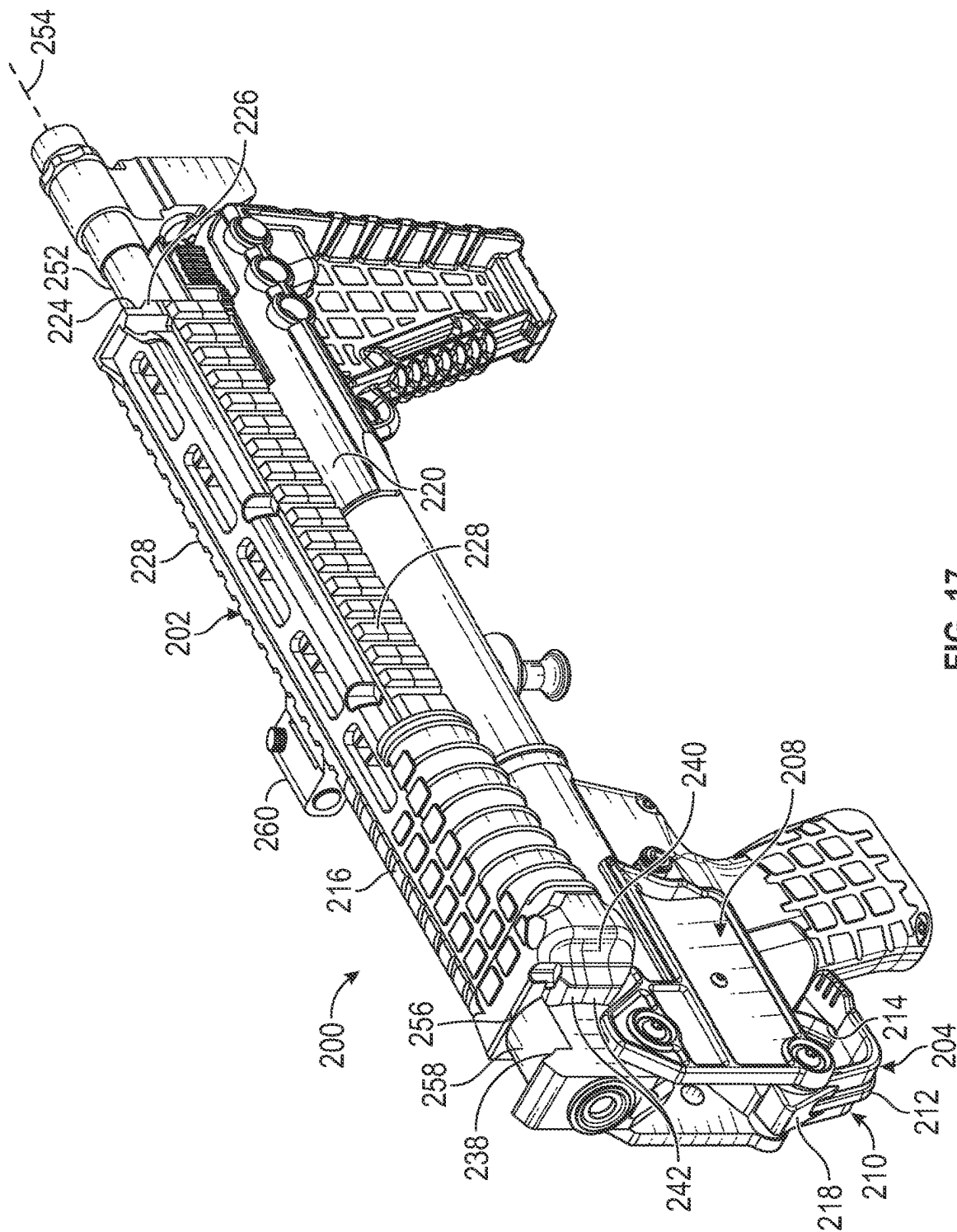


FIG. 17

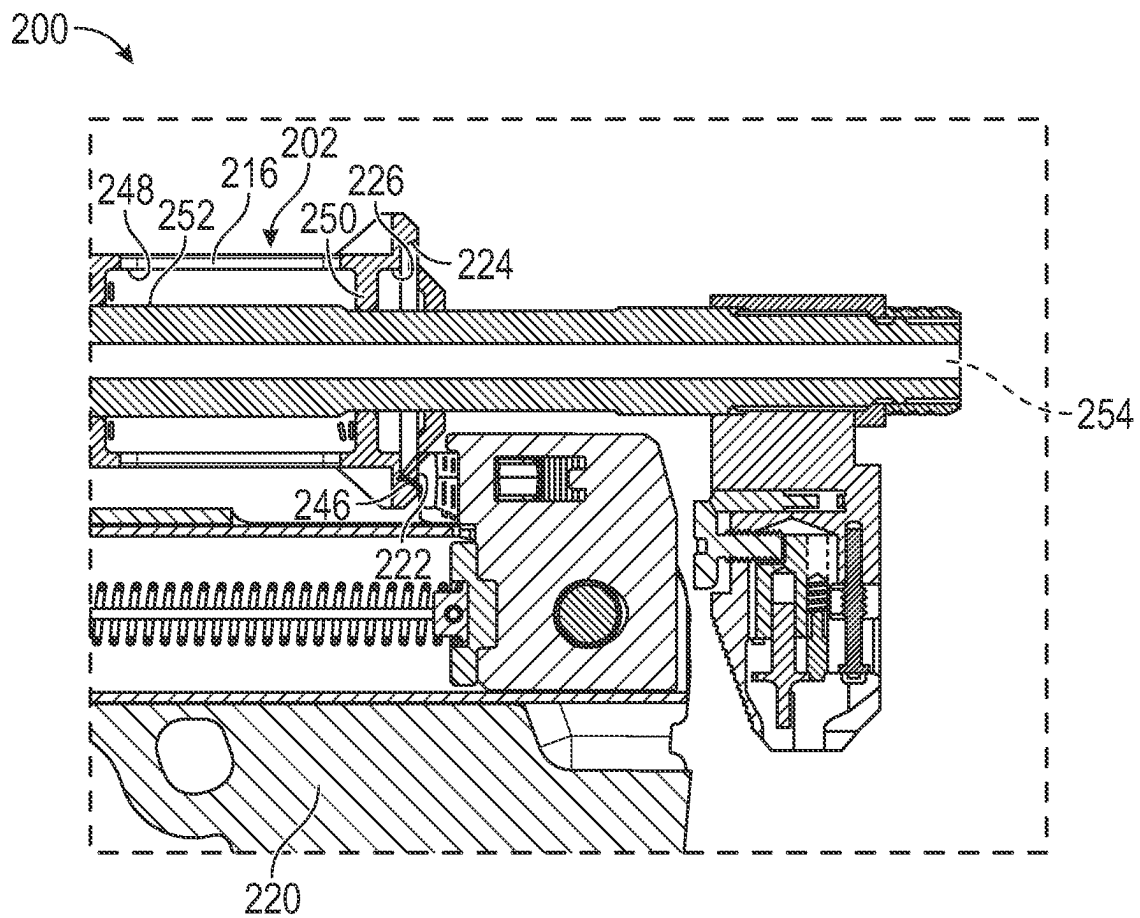


FIG. 18

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FOLDING FIREARM**CROSS-REFERENCE TO RELATED APPLICATION**

This application is a Continuation-in-Part of U.S. patent application Ser. No. 17/146,235 filed on Jan. 11, 2021, entitled "FOLDING FIREARM," which claims the benefit of U.S. Provisional Patent Application No. 62/960,230 filed on Jan. 13, 2020, entitled "Weapon with integrated suppressed barrel with rotating suppressor tube forend to rotate optics and accessories out of the line of sight, to reduce overall height for compact storage," which is hereby incorporated by reference in its entirety for all that is taught and disclosed therein.

FIELD OF THE INVENTION

The present invention relates to firearms, and more particularly to a firearm that has an integrated suppressed barrel with a rotating suppressor tube forend to enable folding despite attached optics.

BACKGROUND AND SUMMARY OF THE INVENTION

Folding firearms are popular for their ability to fold for compact storage. Two prior art examples of such a folding firearm is the SUB-9 and SUB2000 carbine rifles manufactured by KelTec Weapons of Cocoa, FL. They include a folding barrel assembly that folds about a hinge axis to reduce the overall length of the weapon nearly by half. This enables the user to secure the folded firearm in a smaller space as opposed to a larger, rifle-sized gun safe. The folding feature also allows the folded firearm to fit snugly in a small backpack or a briefcase-style container instead of a full-length rifle case for ease of transport.

The disadvantage to these prior art folding firearms is that any mounted optics or added accessories may obstruct folding of the host firearm altogether or limit the extent the host firearm can be folded. Although the mounted optics or accessories could be removed from the weapon prior to folding, this could be frustrating to many users. Not only would the firearm not be immediately ready for use upon unfolding because the items must first be reattached, but mounted optics or other sighting systems might have to be zeroed again to ensure accuracy.

An alternative approach to removing any mounted optics or added accessories prior to folding the firearm is to replace the original factory forend on the SUB2000 carbine rifle with an after-market rotating forend. Examples of rotating forends are the Gen1 and Gen2 Sub 2000 forends manufactured by Red Lion Precision, LLC of Red Lion, PA. The forends rotate in 90° indexed increments, which permits the user to fold the weapon despite optics remaining mounted. The design also enables the user to rotate the optics back into the line of sight for normal firearm operation without disturbing their zero. While this is an effective solution to the problem of not being able to fold the firearm with attached optics or accessories, it is expensive and adds considerable weight to what is otherwise a comparatively inexpensive firearm to purchase and operate. This approach also does not incorporate any suppression capability to reduce the auditory report on weapon discharge, which if added further increases cost, weight, and overall length of the host firearm.

Therefore, a need exists for a new and improved folding firearm that has an integrated suppressed barrel with a

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rotating suppressor tube forend to enable folding despite attached optics. In this regard, the various embodiments of the present invention substantially fulfill at least some of these needs. In this respect, the folding firearm according to the present invention substantially departs from the conventional concepts and designs of the prior art, and in doing so provides an apparatus primarily developed for the purpose of providing a folding firearm having an integrated suppressed barrel with a rotating suppressor tube forend to enable folding despite attached optics.

The present invention provides an improved folding firearm, and overcomes the above-mentioned disadvantages and drawbacks of the prior art. As such, the general purpose of the present invention, which will be described subsequently in greater detail, is to provide an improved folding firearm that has all the advantages of the prior art mentioned above.

To attain this, the preferred embodiment of the present invention essentially comprises a frame having a trigger assembly and defining a bolt passage, a barrel assembly pivotally connected to the frame for movement between an extended operable condition and a folded inoperable condition, the barrel assembly defining a barrel axis, a latch assembly operably connected to the frame and to the barrel assembly, the latch assembly including a latch operable between a first latched condition and a second unlatched condition, when the barrel assembly is in the operable condition and the latch is in the latched condition, the barrel assembly is secured in the operable condition, when the latch is in the unlatched condition, the barrel assembly is free to move away from the operable condition toward the folded condition, and the firearm being free of other restraints on movement of the barrel assembly from the operable condition besides the latch, such that single-control release of the barrel assembly is provided. There are, of course, additional features of the invention that will be described hereinafter and which will form the subject matter of the claims attached.

There has thus been outlined, rather broadly, the more important features of the invention in order that the detailed description thereof that follows may be better understood and in order that the present contribution to the art may be better appreciated.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a top isometric view of the current embodiment of a folding firearm constructed in accordance with the principles of the present invention with the sleeve in the operating condition.

FIG. 2 is a right side view of the folding firearm of FIG. 1 with the sleeve in the operating condition.

FIG. 3 is a top isometric view of the folding firearm of FIG. 1 with the sleeve in the stowage condition.

FIG. 4 is a right side view of the view of the folding firearm of FIG. 1 with the sleeve in the stowage condition.

FIG. 5 is right side view of the folding firearm of FIG. 1 with the sleeve and folding firearm in the stowage condition.

FIG. 6 is an enlarged top isometric view of the folding firearm of FIG. 1 with the latch in the unlatched condition raised above the sleeve.

FIG. 7 is an exploded view of the folding firearm of FIG. 1.

FIG. 8 is a side sectional view of the folding firearm of FIG. 1 with the sleeve in the stowage condition.

FIG. 9 is an enlarged side sectional view of the folding firearm of FIG. 1 showing the trigger guard in the latched and unlatched conditions.

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FIG. 10 is an enlarged side sectional view of the folding firearm of FIG. 1 showing the latch in the unlatched condition lowered onto the sleeve.

FIG. 11 is a right side view of an alternative embodiment of the folding firearm constructed in accordance with the principles of the present invention with the latch in the latched condition and the barrel assembly secured in the operable condition.

FIG. 12A is a side sectional view of the alternative embodiment of the folding firearm of FIG. 11 with the latch in the latched condition and the barrel assembly secured in the operable condition.

FIG. 12B is a fragmentary enlarged sectional isometric view of the alternative embodiment of the folding firearm of FIG. 11 with the latch in the latched condition and the barrel assembly secured in the operable condition.

FIG. 13A is a side sectional view of the alternative embodiment of the folding firearm of FIG. 11 with the latch in the unlatched condition and the barrel assembly free to move away from the operable condition toward the folded condition.

FIG. 13B is a fragmentary enlarged sectional isometric view of the alternative embodiment of the folding firearm of FIG. 11 with the latch in the unlatched condition and the barrel assembly free to move away from the operable condition toward the folded condition.

FIG. 14A is a side sectional view of the alternative embodiment of the folding firearm of FIG. 11 with the latch in the unlatched condition and the barrel assembly pivoted upwards toward the folded condition, which enables rotation of the sleeve.

FIG. 14B is a fragmentary enlarged isometric view of the alternative embodiment of the folding firearm of FIG. 11 with the latch in the unlatched condition and the barrel assembly pivoted upwards toward the folded condition, which enables rotation of the sleeve.

FIG. 15 is a fragmentary isometric view of the barrel assembly of the alternative embodiment of the folding firearm of FIG. 11.

FIG. 16 is an isometric view of the alternative embodiment of the folding firearm of FIG. 11 with the barrel assembly in the folded condition and the sleeve in the first orientation, which is non-rotated.

FIG. 17 is an isometric view of the alternative embodiment of the folding firearm of FIG. 11 with the barrel assembly in the folding condition and the sleeve in the second orientation, which is rotated 90° from the first orientation.

FIG. 18 is a fragmentary side sectional view of the alternative embodiment of the folding firearm of FIG. 11 with the barrel assembly in the folded condition and the sleeve in the second orientation, which is rotated 90° from the first orientation.

The same reference numerals refer to the same parts throughout the various figures.

DESCRIPTION OF THE CURRENT EMBODIMENT

An embodiment of the folding firearm of the present invention is shown and generally designated by the reference numeral 10.

FIGS. 1-5, 7, and 8 illustrate the improved folding firearm 10 of the present invention. More particularly, FIGS. 1 and 2 show the folding firearm in the unfolded condition with the sleeve 12 in the operating condition, FIGS. 3 and 4 show the folding firearm in the unfolded condition with the sleeve in

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the stowage condition, and FIG. 5 shows the folding firearm in the folded condition with the sleeve in the stowage condition. The folding firearm has a frame 14 with a grip 16 and a stock 18 attached to the rear 122. The frame also has a trigger assembly 20 and defines a passage 22 containing a bolt. A barrel assembly 26 is pivotally connected to the frame by a hinge 28 that serves as a pivot facility to move between an operating condition in which the barrel assembly extends away from the frame and a folded condition in which the barrel assembly is adjacent to the frame. The barrel assembly includes a barrel element 30 defining a barrel axis 32. The barrel element includes a barrel 34, a plurality of steel suppressor baffles 36 forward of the barrel, a plurality of aluminum suppressor baffles 38 forward of the plurality of steel suppressor baffles, and an interior space 40. The barrel assembly also includes the sleeve, which encompasses the barrel. The sleeve has an interior 42 defining an expansion chamber 44 in gas communication with the interior space of the barrel element. A front closure 46 and a rear closure 48 at respective ends of the sleeve (front 50 and rear 52) provide a gas seal enclosing the expansion chamber. In the current embodiment, the sleeve is rotatable about the barrel axis and includes a mounting rail 58 formed by a top 56 of the sleeve capable of receiving any compatible sight facility 54. Optionally, a bottom 58 of the sleeve forms an additional mounting rail 60.

In the current embodiment, the sleeve 12 is movable between an operating condition in which the folding firearm 10 is operable and the sight facility 54 is in a first position and a stowage condition in which the sight facility is rotationally offset from the first position. The amount of rotational offset of the sight facility from the first position can be 90° clockwise or counterclockwise in the current embodiment. The first position is on a medial plane 62 of the folding firearm. The sleeve is rotatable about the barrel axis such that the sight facility may be rotated aside to enable movement of the barrel assembly 26 to the folded condition depicted in FIG. 5. The hinge 28 includes a pivotally connected latch 64 that is operable to selectively secure the sleeve in the operating condition by engaging a latch pocket 66 defined by the top 56 of the sleeve with a forward portion 68. The interaction between the forward portion of the latch and the latch pocket ensures precise, repeatable alignment of the sight facility with the frame 14 each time the sleeve is returned to the operating condition from the stowage condition. As a result, the sight facility does not have to be zeroed again to restore accuracy to the folding firearm 10.

The barrel 34 includes a rifled portion 70. The barrel element 30 includes an inner tube 72 encompassing the plurality of steel suppressor baffles 36 and aluminum suppressor baffles 38. The inner tube also includes a plurality of rear end forward gas apertures 74, 76 providing gas communication between the interior space 40 of the barrel element, which is defined by the inner tube, and the expansion chamber 44 defined by the interior 42 of the sleeve 12. In the current embodiment, the sleeve is a single wall body, includes a mounting rail (mounting rail 58), is a unitary body, and is free of openings except at its front and rear ends 50, 52.

The hinge 28 has a top 78, bottom 80, front 82, and rear 84. The top of the hinge includes left and right ears 86, 88, each of which defines an aperture 90, 92 to enable the latch 64 to be pivotally connected to the hinge by a screw 94. The hinge also defines a central bore 96 that extends from the front to the rear. The rear 96 of the barrel 34 threads into the central bore, which allows the barrel assembly 26 to rotate about the barrel axis 32.

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A left hand guard **98** having an interior **100** and an exterior **102** and a right hand guard **104** having an interior **106** and an exterior **108** have their interiors clamped against the exterior **110** of the sleeve **12** by a plurality of screws **112** threadedly received by a plurality of locking nuts **114**. In the current embodiment, the left and right hand guards are made of polymer to prevent undesirable heat transfer from the sleeve to the user's hands. A bevel washer **116** presses against a washer **118** and the rear closure **48**. This causes the washer to press against the front **120** of the frame **14** for tension when the folding firearm **10** is in the operating condition and presses the barrel assembly **26** into the trigger guard **124**.

The rear closure **48** is a forend nut that has a rear **126** screwed into the hinge **28**. The rear closure locks the barrel **34** in place so the barrel will not unthread from the hinge during rotation between the operating and stowage conditions. The rear closure also has threads on the front **128** that engage threads **130** in the rear **52** interior **42** of the sleeve **12** to enable the sleeve to rotate clockwise and counterclockwise about the barrel axis **32**. The rear closure also serves as a gas seal for the rear of the expansion chamber **44**. A central bore **132** that receives the rear **96** of the barrel is defined by the rear closure, and a flange **134** is located between the front and rear of the rear closure. An O-ring **136** assists the rear closure in creating the gas seal for the rear of the expansion chamber. The O-ring also creates intentional interference between the internal diameter of the rear interior of the sleeve and the front of the rear closure to remove any slack between the threads on the front of the rear closure and the rear interior of the sleeve.

The rear **96** of the barrel **34** defines a central bore **138** having a rifled portion **70** that is about 4.25 inches long in the current embodiment. The barrel is connected to the rear **140** of the inner tube **74**. The exterior **142** of the inner tube defines a plurality of rear gas apertures **74** and forward gas apertures **76** that allow gases resulting from the discharge of the folding firearm **10** to vent into the expansion chamber **44**. The front **144** of the inner tube threadedly receives a front closure **46** that serves as a gas seal for the front **50** of the expansion chamber. The front closure defines a central bore **146** that permits a bullet to exit the inner tube.

The interior space of the barrel assembly **26** defined by the inner tube **72** receives the plurality of steel suppressor baffles **36** and the plurality of aluminum suppressor baffles **38**. The steel suppressor baffles have a front **148**, rear **150**, exterior **152**, and define a plurality of gas passages **154** that communicate between the exterior and a central bore **154**. There are four steel suppressor baffles in the current embodiment. The steel suppressor baffles are the rearmost suppressor baffles and are hardened to withstand the initial blast of gas pressure resulting from discharge of the folding firearm **10**. The bullet passes through the central bores of the steel suppressor baffles, and the angled cone shape of the steel suppressor baffles assists in venting gases away from the traveling bullet. The steel suppressor baffles can also be made of other suitable materials capable of withstanding the operating environment. The aluminum suppressor baffles have a front **156**, rear **158**, exterior **160**, and define a plurality of gas passages **162** that communicate between the exterior and a central bore **164**. The aluminum suppressor baffles are located forward of the steel suppressor baffles and are subjected to reduced gas temperature and pressure compared to the steel suppressor baffles. The bullet passes through the central bores of the aluminum suppressor baffles, and the angled cone shape of the aluminum suppressor baffles assists in venting gases away from the

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traveling bullet. The aluminum suppressor baffles are longer than the steel suppressor baffles in the current embodiment and can be made of other suitable materials capable of withstanding the operating environment.

FIGS. **6** and **10** illustrate the improved folding firearm **10** of the present invention. More particularly, the latch **64** is shown with the forward portion **68** disengaged from the latch pocket **66**. With the latch in the illustrated positions, the sleeve is free to rotate clockwise or counterclockwise about the barrel axis **32** to change the sight facility between the operating condition and the stowage condition. When the sleeve is in the operating condition, the forward portion of the latch is received by the latch pocket to prevent sleeve rotation. When the sleeve is in the stowage condition, the forward portion of the latch rests against the exterior **110** of the sleeve and does not prevent sleeve rotation.

FIG. **9** illustrates the improved folding firearm **10** of the present invention. More particularly, the trigger guard **124** that encircles the trigger **166** is pivotally connected to the bottom **168** front **120** of the frame **14** by a screw **170**. The trigger guard pivots between the two positions shown to lock and unlock the hinge **28** from the frame. In the locked position, the trigger guard is positioned such that a forward portion **172** engages a step **174** that protrudes from the bottom **80** rear **84** of the hinge and a rearward portion **176** contacts the grip **16**. In the unlocked position, the user has rotated the trigger guard clockwise such that the forward portion of the trigger guard disengages from and no longer obstructs the step on the hinge to permit the hinge to rotate about the screw **94** to transition the barrel assembly **26** from the operating condition to the folded position. Left and right ears **86**, **88** on the hinge are received between left and right ears **178**, **180** on the top **182** of the frame that each define an aperture **184**, **186** that receives the screw. When the barrel assembly is in the operating condition, it should be appreciated that the hinge is received within the front **120** of the passage **22** in the frame.

FIGS. **11-14B** and **16-18** illustrate an alternative embodiment of the improved folding firearm **200** of the present invention. FIG. **15** illustrates a forward portion of a barrel assembly **202**. More particularly, the folding firearm **200** has a frame **208** having a trigger assembly **204** and defines a bolt passage **206**. The barrel assembly is pivotally connected to the frame for movement between an extended operable condition shown in FIGS. **11-12B** and a folded inoperable condition shown in FIGS. **16-18**. A latch assembly **210** is operably connected to the frame and to the barrel assembly. The latch assembly includes a latch **212** operable between a first latched condition shown in FIGS. **11-12B** and a second unlatched condition shown in FIGS. **13A-14B**. When the latch is in the unlatched condition, the barrel assembly is free to move away from the operable condition toward the folded condition, which is shown in FIGS. **13A-14B**. The folding firearm **200** is free of other restraints on movement of the barrel assembly from the operable condition besides the latch, such that single-controlled release of the barrel assembly is provided. Thus, barrel assembly is enabled to move from the operable condition to the stowed condition by a single movement of the latch assembly. In the current embodiment, the latch **212** is pivotally connected to the frame **208** and is a trigger guard associated with the trigger assembly **204**. The latch is configured to guard a trigger lever **214** of the trigger assembly when the latch is in the latched condition.

The barrel assembly **202** includes a rotatable sleeve **216** operable to rotate between a first orientation shown in FIGS. **11-16** and a second orientation shown in FIGS. **17-18**. In the

current embodiment, the first and second orientations are 90° offset from each other, which enables the barrel assembly to be folded even when an optic **260** or another accessory is attached to one of the mounting rails **228** on the sleeve. Alternatively, the sleeve could be hexagonal, octagonal, or any other suitable shape in cross-section with any suitable quantity of latching areas at any suitable offset from each other, such as 60° or 45°. Rotation of the sleeve is enabled by a series of pockets **248** that create space between the sleeve and the barrel **252**. A series of posts **250** contact the barrel and ensure the sleeve remains centered while rotating about the barrel axis **254**. It should be appreciated that the rear of the sleeve has a rounded portion **256** that matches a rounded area **258** on the hinge to enable the sleeve to rotate. The latch **212** includes a rotation block element **218** operable to secure the sleeve in the first orientation when the barrel assembly is in the operable condition and the latch is in the latched condition. Protrusions from the forend **242** also assist with preventing the sleeve from rotating while in the operational condition. The rotation block element is configured to enable rotation of the sleeve when the latch is in the unlatched condition.

The frame **208** includes an adjustable buttstock **220**. The barrel assembly **202** is configured to overlay the buttstock when in the folded condition (shown in FIGS. **16-18**). The sleeve **216** includes a first restraint element **222** operable to secure the barrel assembly in the folded condition when the sleeve is in the first orientation. The sleeve also includes a second restraint element **224** operable to secure the barrel assembly in the folded condition when the sleeve is in the second orientation (visible in FIG. **15**). In the current embodiment, each of the first and second restraints are at forward ends **226** of the sleeve and each of the first and second restraints are associated with mounting rails **228** on the sleeve. The top rear of the folding firearm **10** includes a spring-biased latch **246** connected to a buffer. The spring-biased latch engages with whichever of the first and second restraints is aligned with the spring-biased latch when the barrel assembly is in the folded condition to releasably secure the barrel assembly in the folded condition.

The latch **212** includes a trigger block **230** configured to enable operation of the trigger assembly when the latch is in the latched condition. The barrel assembly **202** is pivotally connected to the frame **208** to pivot on a barrel pivot axis **232**. The latch is pivotally connected to pivot on a latch pivot axis **234**. In the current embodiment, barrel pivot axis and latch pivot axis are parallel to each other. The barrel pivot axis and latch pivot axis are transverse to a major vertical plane **236** defined by the firearm.

It should be appreciated that when the latch **212** is in the latched condition, the trigger block **230** contacts a curved hinge area **238** to prevent the barrel assembly **202** from pivoting out of the operable condition and to wedge the latch into the latched condition. The latch and trigger block act as a trigger guard to guard the trigger lever **214**, lock the barrel assembly in place to prevent the barrel assembly from moving, and assist in locking the sleeve in place **216** to prevent the sleeve from rotating in the operational state. The rotation block element **218** is received in the forwardmost portion of a pocket **240** defined by the bottom rear of the sleeve **216**, and feet **242** protruding rearwardly from the barrel assembly into the frame **208**, prevent the sleeve from rotating while the barrel assembly is in the operable condition. When the latch is in the unlatched condition, the latch has pivoted counterclockwise from the latched condition such that the rotation block element is received in the rearward most portion of the pocket. The trigger block no

longer contacts the hinge area when the latch is in the unlatched condition. When the latch is in the unlatched condition, the barrel assembly can pivot about the hinge **244**, which enables the feet to withdraw from the frame and the pocket to release the rotation block element. The shape of the rotation block element is determined by the interactions with the pocket to prevent rotation of the sleeve in the forwardmost portion of the pocket and to enable the pocket to release the rotation block element in the rearwardmost portion of the pocket.

While current embodiments of a folding firearm have been described in detail, it should be apparent that modifications and variations thereto are possible, all of which fall within the true spirit and scope of the invention. For example, the folding firearm could be a pistol with a brace instead of a rifle with a buttstock and buffer as described. With respect to the above description then, it is to be realized that the optimum dimensional relationships for the parts of the invention, to include variations in size, materials, shape, form, function and manner of operation, assembly and use, are deemed readily apparent and obvious to one skilled in the art, and all equivalent relationships to those illustrated in the drawings and described in the specification are intended to be encompassed by the present invention.

Therefore, the foregoing is considered as illustrative only of the principles of the invention. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the invention to the exact construction and operation shown and described, and accordingly, all suitable modifications and equivalents may be resorted to, falling within the scope of the invention.

I claim:

1. A firearm comprising:

- a frame having a trigger assembly and defining a bolt passage;
- a barrel assembly pivotally connected to the frame for movement between an extended operable condition and a folded inoperable condition;
- the barrel assembly defining a barrel axis;
- a latch assembly operably connected to the frame and to the barrel assembly;
- the latch assembly including a latch operable between a first latched condition and a second unlatched condition;
- when the barrel assembly is in the operable condition and the latch is in the latched condition, the barrel assembly is secured in the operable condition;
- when the latch is in the unlatched condition, the barrel assembly is free to move away from the operable condition toward the folded condition;
- the firearm being free of other restraints on movement of the barrel assembly from the operable condition besides the latch, such that single-control release of the barrel assembly is provided; and
- wherein the latch is a trigger guard associated with the trigger assembly and configured to guard a trigger lever of the trigger assembly when the latch is in the latched condition.

2. The firearm of claim **1** wherein the latch is pivotally connected to the frame.

3. The firearm of claim **1** wherein the first and second orientations are 90° offset from each other.

4. The firearm of claim **3** wherein each of the first and second restraints are at forward ends of the sleeve.

5. The firearm of claim **3** wherein each of the first and second restraints are associated with mounting rails on the sleeve.

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6. The firearm of claim 1 wherein the frame includes a buttstock and wherein the barrel assembly is configured to overlay the buttstock when in the folded condition, and wherein the sleeve includes a first restraint element operable to secure the barrel assembly in the folded condition when the sleeve is in the first orientation, and the sleeve includes a second restraint element operable to secure the barrel assembly in the folded condition when the sleeve is in the second orientation.

7. The firearm of claim 1 wherein the barrel assembly is pivotally connected to the frame to pivot on a barrel pivot axis and the latch is pivotally connected to pivot on a latch pivot axis, and the barrel pivot axis and latch pivot axis are parallel to each other.

8. The firearm of claim 7 wherein the barrel pivot axis and latch pivot axis are transverse to a major vertical plane defined by the firearm.

9. The firearm of claim 1 wherein the barrel assembly is enabled to move from the operable condition to the stowed condition by a single movement of the latch assembly.

10. A firearm comprising:

a frame having a trigger assembly and defining a bolt passage;

a barrel assembly pivotally connected to the frame for movement between an extended operable condition and a folded inoperable condition;

the barrel assembly defining a barrel axis;

a latch assembly operably connected to the frame and to the barrel assembly;

the latch assembly including a latch operable between a first latched condition and a second unlatched condition;

when the barrel assembly is in the operable condition and the latch is in the latched condition, the barrel assembly is secured in the operable condition;

when the latch is in the unlatched condition, the barrel assembly is free to move away from the operable condition toward the folded condition;

the firearm being free of other restraints on movement of the barrel assembly from the operable condition besides the latch, such that single-control release of the barrel assembly is provided; and

wherein the latch includes a trigger block configured to enable operation of the trigger assembly when the latch is in the latched condition.

11. The firearm of claim 10 wherein the latch is pivotally connected to frame.

12. The firearm of claim 10 wherein the first and second orientations are 90° offset from each other.

13. The firearm of claim 12 wherein each of the first and second restraints are at forward ends of the sleeve.

14. The firearm of claim 12 wherein each of the first and second restraints are associated with mounting rails on the sleeve.

15. The firearm of claim 10 wherein the frame includes a buttstock and wherein the barrel assembly is configured to overlay the buttstock when in the folded condition, and wherein the sleeve includes a first restraint element operable to secure the barrel assembly in the folded condition when the sleeve is in the first orientation, and the sleeve includes a second restraint element operable to secure the barrel assembly in the folded condition when the sleeve is in the second orientation.

16. The firearm of claim 10 wherein the barrel assembly is pivotally connected to the frame to pivot on a barrel pivot

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axis and the latch is pivotally connected to pivot on a latch pivot axis, and the barrel pivot axis and latch pivot axis are parallel to each other.

17. The firearm of claim 16 wherein the barrel pivot axis and latch pivot axis are transverse to a major vertical plane defined by the firearm.

18. The firearm of claim 10 wherein the barrel assembly is enabled to move from the operable condition to the stowed condition by a single movement of the latch assembly.

19. A firearm comprising:

a frame having a trigger assembly and defining a bolt passage;

a barrel assembly pivotally connected to the frame for movement between an extended operable condition and a folded inoperable condition;

the barrel assembly defining a barrel axis;

a latch assembly operably connected to the frame and to the barrel assembly;

the latch assembly including a latch operable between a first latched condition and a second unlatched condition;

when the barrel assembly is in the operable condition and the latch is in the latched condition, the barrel assembly is secured in the operable condition;

when the latch is in the unlatched condition, the barrel assembly is free to move away from the operable condition toward the folded condition;

wherein the barrel assembly includes a rotatable sleeve encompassing the barrel and operable to rotate about the barrel between a first orientation and a second orientation; and

wherein the latch includes a rotation block element operable to secure the sleeve in the first orientation when the barrel assembly is in the operable condition and the latch is in the latched condition.

20. The firearm of claim 19 wherein the latch is pivotally connected to frame.

21. The firearm of claim 19 wherein the first and second orientations are 90° offset from each other.

22. The firearm of claim 21 wherein each of the first and second restraints are at forward ends of the sleeve.

23. The firearm of claim 21 wherein each of the first and second restraints are associated with mounting rails on the sleeve.

24. The firearm of claim 19 wherein the frame includes a buttstock and wherein the barrel assembly is configured to overlay the buttstock when in the folded condition, and wherein the sleeve includes a first restraint element operable to secure the barrel assembly in the folded condition when the sleeve is in the first orientation, and the sleeve includes a second restraint element operable to secure the barrel assembly in the folded condition when the sleeve is in the second orientation.

25. The firearm of claim 19 wherein the barrel assembly is pivotally connected to the frame to pivot on a barrel pivot axis and the latch is pivotally connected to pivot on a latch pivot axis, and the barrel pivot axis and latch pivot axis are parallel to each other.

26. The firearm of claim 25 wherein the barrel pivot axis and latch pivot axis are transverse to a major vertical plane defined by the firearm.

27. The firearm of claim 19 wherein the barrel assembly is enabled to move from the operable condition to the stowed condition by a single movement of the latch assembly.

28. The firearm of claim 19 including the firearm being free of other restraints on movement of the barrel assembly

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from the operable condition besides the latch, such that single-control release of the barrel assembly is provided.

29. A firearm comprising:

a frame having a trigger assembly and defining a bolt passage;

a barrel assembly pivotally connected to the frame for movement between an extended operable condition and a folded inoperable condition;

the barrel assembly defining a barrel axis;

a latch assembly operably connected to the frame and to the barrel assembly;

the latch assembly including a latch operable between a first latched condition and a second unlatched condition;

when the barrel assembly is in the operable condition and the latch is in the latched condition, the barrel assembly is secured in the operable condition;

when the latch is in the unlatched condition, the barrel assembly is free to move away from the operable condition toward the folded condition; and

wherein the barrel assembly includes a rotatable sleeve encompassing the barrel and operable to rotate about the barrel between a first orientation and a second orientation, and wherein the latch includes a rotation block element operable to enable rotation of the sleeve when the latch is in the unlatched condition.

30. The firearm of claim **29** wherein the latch is pivotally connected to frame.

31. The firearm of claim **29** wherein the first and second orientations are 90° offset from each other.

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32. The firearm of claim **31** wherein each of the first and second restraints are at forward ends of the sleeve.

33. The firearm of claim **31** wherein each of the first and second restraints are associated with mounting rails on the sleeve.

34. The firearm of claim **29** wherein the frame includes a buttstock and wherein the barrel assembly is configured to overlay the buttstock when in the folded condition, and wherein the sleeve includes a first restraint element operable to secure the barrel assembly in the folded condition when the sleeve is in the first orientation, and the sleeve includes a second restraint element operable to secure the barrel assembly in the folded condition when the sleeve is in the second orientation.

35. The firearm of claim **29** wherein the barrel assembly is pivotally connected to the frame to pivot on a barrel pivot axis and the latch is pivotally connected to pivot on a latch pivot axis, and the barrel pivot axis and latch pivot axis are parallel to each other.

36. The firearm of claim **35** wherein the barrel pivot axis and latch pivot axis are transverse to a major vertical plane defined by the firearm.

37. The firearm of claim **29** wherein the barrel assembly is enabled to move from the operable condition to the stowed condition by a single movement of the latch assembly.

38. The firearm of claim **29** including the firearm being free of other restraints on movement of the barrel assembly from the operable condition besides the latch, such that single-control release of the barrel assembly is provided.

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